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% 9/25/25
% function to calculate C and S (gets input to laguerre)

function [C, S] = calcCandS(y)
    if y > 0 % elliptical
        C = (1 - cos(sqrt(y))) / y;
        S = (sqrt(y) - sin(sqrt(y))) / (sqrt(y^3));
    elseif y < 0 % hyperbolic
        C = (cosh(sqrt(-y))-1) / -y;
        S = (sinh(sqrt(-y)) - sqrt(-y)) / (sqrt(-y^3));
    else
        C = 1; % idrk
        S = 1;
    end
end
```